

Arika Khor

SWE & Researcher with Interest in Autonomous Systems, Digital Facial Privacy Protection, and System Design

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Education

Expected B.S. 2026, M.S. 2027	The University of Oklahoma B.S. / M.S. Computer Science, focus in Robotics & Computer Vision
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Experience

June 2026 - Current	The Boeing Company <i>Intern: Simulation Software Engineering</i> · Containerized legacy on-premises software with Podman to optimize development and local testing workflows. · Engineered a multi-agent Retrieval-Augmented Generation (RAG) framework with SQLite to automate Software Design Document creation. · Maintained CI/CD container pipelines to ensure stable delivery to Kubernetes environments. · Developed knowledge graphs in Obsidian to streamline system mapping and data traceability.
June 2025 - Current	Construction Innovation and Analytics Lab (CinnaLAB) <i>Time to Incident Predictor on an Autonomous Vehicle in Dynamic Construction Sites</i> · Deployed an autonomous ground vehicle for navigation in active, high-risk construction environments. · Integrated LIDAR, stereo depth cameras, and GNSS to refine real-time localization and spatial mapping. · Outlined technical systems design architecture for predictive incident analysis tools monitoring site hazards.
Aug 2025 - Current	Autonomous Interactive Robotics Lab (AIROU) <i>Collaborative Autonomous Vehicles at Smart Intersections with Nvidia Isaac Lab</i> · Engineered high-fidelity reinforcement learning simulations evaluating automated vehicle behavior and V2I communication logic. · Formulated automated policy correction frameworks to validate decision-making routines without manual intervention. <i>ARCPro Autonomous Model Race Cars</i> · Deployed Ansible configuration pipelines to standardize system setup across robotics hardware. · Developed user interface web applications and technical documentation portals for system monitoring. · Built an autonomous software stack featuring sensor fusion and SLAM capabilities.

Notable Personal Projects

2025 Hacklahoma	Real Time Local Facial Recognition · Engineered a localized real-time facial recognition application utilizing MongoDB for secure identity tracking. · Constructed intuitive web control interfaces using Streamlit and Flask to manage identity profiles.
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School Based Organizations

Aug 2024 - Current	Sooner Competitive Robotics <i>Software Member & Treasurer (25-26)</i> · Executed fundamental PCB layouts in Altium and structural CAD modeling in Onshape for introductory vehicle development kits. · Contributed to cross-functional engineering teams utilizing ROS2 publish-subscribe architectures to implement cohesive vehicle system designs.
Aug 2016 - May 2022	Scouting America <i>Various Leadership Positions</i> · Eagle Scout Project: Directed team workflows to mass-produce and distribute emergency weather radio antennas for public service utilization.
Aug 2024 - Current	Sooner Competitive Programming Team <i>Team Captain (24-26, Team Advisor (26-27)</i>

Technical Skills

Core Competencies · Systems Design, Autonomous Systems, Program Coordination, Cross-Functional Teamwork, Technical Documentation, Agile/Scrum.	Programming Languages · Java (Spring, Thymeleaf), C++ , C#, JavaScript, Python, Bash.
Tools & Infrastructure · Docker, Podman, Kubernetes, Ansible, Git, Jira, Jenkins, Tailscale, Linux Ubuntu, Debian.	Frameworks & Databases · ROS2, SQL, MongoDB.